

Key Outcomes

Team finalized backend architecture using **Postgres/Supabase** with **FastAPI** and **UV** package manager, completed mobile/web mockups in Figma, and developed user questionnaire to validate product features including busyness prediction and spontaneous event discovery

Decisions Made

- **Backend stack:** Postgres database via Supabase (500MB free tier), FastAPI framework, UV for dependency management instead of pip
- **Migration tool:** Alembic tentatively selected for SQL Alchemy migrations, pending final confirmation
- **Data strategy:** Use real event data from NYC Open Data; synthesize user data when needed for simulation
- **Recommendation logic:** Prioritize nearest events, supplemented by user saved preferences and attendance metrics rather than ratings
- **Accessibility scope:** Focus on transportation accessibility features rather than venue-specific wheelchair access
- **Rating system:** Deferred decision on 1-5 star ratings due to concerns about fake reviews; consider alternative metrics like monthly organizer attendance

Pending Confirmation

- **Alembic migration tool:** Needs validation whether required given Supabase setup
- **Event data sources:** Final selection between Ticketmaster API, NYC Open Data historical events, and Mayor's office summer activities
- **Busyness model anchoring:** Decision needed on whether to base predictions on subway stations, bike stations, or taxi zones
- **Testing software stack:** Team needs to confirm preferred tools for front-end (Jest for JavaScript) and back-end performance/security testing
- **Mobile deployment:** Verify cost and feasibility of QR code vs app store vs mini-program approach

Questionnaire Feedback & Revisions

Additions required by Thursday evening deadline:

- **Busyness section:** Questions on how crowdedness affects decision to attend events, preference for busy vs quiet venues, concerns about getting home from crowded areas
- **Spontaneity section:** Questions validating last-minute event attendance vs advance planning preferences
- **Transportation in busyness context:** Expand beyond cost to include impact of crowds on travel duration and mode choice

Current questionnaire covers: Demographics, price sensitivity, activity motivation, transportation costs/preferences, interests (food/music/sports/nightlife), AI summary

usefulness, accessibility needs, grocery shopping habits (Too Good To Go experience), weekly deals interest, overall app concept validation

Rationale clarifications:

- Grocery/food questions stem from user interviews revealing preference for discounted fresh produce over bakery items from Too Good To Go
- Questions validate whether target students prioritize food discounts over event tickets

Development Progress

Backend (Dan):

- Created database schema with minimal redundancy, including radius field for busyness heat map areas
- Architecture documented in `architecture.md` with ER diagrams
- Will push to GitHub for team review, grant admin access to Supabase for all members
- Next: Create API endpoints, integrate Leaflet maps with route calculation

UI/UX (Xiya):

- Completed Figma mockups for web partner dashboard (listings, bookings, revenue analytics, search filters) and mobile app (heat map, category browsing, distance/price filters, user preferences, ratings)
- Will generate design specifications and reference code later from Figma for front-end team
- Next: Finalize questionnaire, create Gantt chart for sprint planning

Machine Learning (Erwan):

- Focused on busyness prediction model using historical subway/taxi/bike data (3M+ records/month) combined with weather and events data
- Working with 2025-present data window to keep datasets manageable
- Discovered NYC Mayor's office summer activities dataset targeting young people—validates market need and provides additional event data
- Next: Finalize dataset merging approach (zones vs stations), complete testing plan assignment

Front-end Web (Sayli):

- Selected **React** framework for web application development
- Next: Review backend endpoints and database schema once pushed, build initial web app mock based on Figma designs

Front-end Mobile (Kezhen):

- Learning **Flutter/Dart** with declarative UI framework approach
- Next: Implement mobile app once Dan creates backend folder structure in GitHub

Action Items

- **Dan:** Push backend structure to GitHub today, share Supabase database access with team, add busyness/spontaneity questions to questionnaire by Thursday evening

- **All team members:** Review [architecture.md](#) and provide feedback by Friday; finalize questionnaire additions by Thursday evening; update timesheets before Friday sprint review
- **Xiya:** Share Figma design specs; develop the Gantt chart next week; attend Friday meeting remotely
- **Erwan:** Share testing plan with team (excluding WHO column until workload confirmed); specify input/output features for busyness model; validate database schema fits ML pipeline needs
- **Sayli & Kezhen:** Review backend endpoints and database structure once available; confirm testing tool preferences for front-end stack
- **Team:** Distribute finalized questionnaire to friends/student groups; prepare sprint review materials for Friday (updated team contract, business plan, timesheets, development status)

Sprint Review Preparation (Friday)

Materials to present: Updated team contract, business plan, timesheets, backlogs, current development status

Xiya attending remotely due to other appointment

Technical Architecture Highlights

Database design: Includes user preferences, event listings with AI-generated summaries, busyness areas with radius fields, organizer analytics, saved events for recommendation engine

Data volume estimates: Can support 2,000 users and 3,000 venues/events within 500MB Supabase free tier; ML training datasets too large for online storage but trained models will fit

Busyness prediction approach: Hourly busyness scores per area using past subway/taxi/bike activity, weather conditions, and concurrent events to predict future crowdedness

Event data richness: NYC Open Data contains 40M+ historical event records since 2010; Ticketmaster provides current listings; Mayor's office offers curated youth activities